VANTIS
ENERGY SOLUTIONS

Knowledge Base

AI Engineer Round 2 — Technical Assignment

Document Type	Customer & Sales Knowledge Base
Version	2.0
Purpose	Source of truth for the AI Sales & Lead Qualification Agent
Last Updated	October 2026

Participant Notice: This document is the primary knowledge source for the assignment. Candidates should build their RAG system around the information contained here and must not assume facts that are not supported by this document.

1. Company Overview

Vantis Energy Solutions is a solar energy solutions company focused on helping residential and small commercial customers reduce electricity costs through rooftop solar installations.

Vantis provides end-to-end solar solutions including solar consultation, rooftop installation, system sizing, residential and small commercial solutions, panel and inverter installation, system monitoring, maintenance support, and post-installation service.

Source-of-truth rule: The AI must not invent product specifications, pricing, subsidies, savings, installation timelines, or service availability that are not present in this knowledge base.

2. Customer Segments

Residential

Individual homeowners, independent houses, villas, and small residential properties. Typical needs include reducing electricity bills, rooftop solar installation, system sizing, installation cost, maintenance, and solar education.

Small Commercial

Shops, offices, clinics, small warehouses, and small business facilities. Commercial evaluation considers electricity consumption, property characteristics, rooftop area, location, and project requirements.

3. Solar Solutions

Recommended system size depends on electricity consumption, monthly bill, property type, rooftop area, customer requirements, location, and budget.

The AI should **not recommend a specific system size** without sufficient customer information. For sizing questions, collect relevant information before making a recommendation.

4. Residential & Commercial Requirements

Residential consultations may consider monthly electricity bill, average consumption, property type, rooftop availability, location, desired timeline, and budget.

Commercial consultations may additionally require business type, property ownership/rental status, approximate project size, and expected installation timeline.

5. Pricing

Vantis does not maintain a single fixed public price for all installations. Cost depends on system size, equipment, installation requirements, property characteristics, location, and project complexity.

Never fabricate an exact project price. Explain that a consultation is required to determine project-specific cost.

6. Installation & Site Assessment

General process: Customer Inquiry → Requirement Assessment → Site/Requirement Evaluation → System Recommendation → Commercial Discussion → Installation Planning → Installation → System Handover.

Installation timing depends on project requirements, site conditions, equipment availability, documentation, and scheduling. A site assessment may evaluate rooftop availability, installation area, property conditions, electrical requirements, and other project-specific considerations.

Do not guarantee an installation date unless it is explicitly provided in the knowledge base.

7. Service Areas

Vantis currently focuses on selected locations in North India. Installation availability varies by location.

If a location is not listed in this document, the AI must not claim that Vantis operates there. It should collect the location and offer verification by the sales team.

8. Product & Service Catalog

The following catalog contains specific fictional Vantis offerings for this assessment. These entries are intended to be retrievable by the AI and should be treated as structured company information.

Offering	Primary Customer	Reference Profile	Important Notes
Vantis HomeStart 3	Residential	3 kW rooftop solution for moderate household electricity use	Final suitability depends on consumption, rooftop area and site assessment.
Vantis HomePro 5	Residential	5 kW rooftop solution for higher household electricity use	Designed as a residential reference package; final configuration is project-specific.
Vantis HomeMax 10	Residential	10 kW rooftop solution for large homes / high consumption	Requires suitable rooftop area and project assessment.
Vantis Business 15	Small Commercial	15 kW reference solution for shops, offices and small facilities	Commercial assessment required before final recommendation.
Vantis Business 25	Small Commercial	25 kW reference solution for larger small-business requirements	Final configuration depends on load, rooftop and site conditions.
Vantis Site Assessment	Residential & Commercial	Pre-installation evaluation	Used to assess rooftop, property and installation requirements.
Vantis Care	Existing Customers	Post-installation support and maintenance assistance	Specific support scope depends on project/equipment terms.

Important Catalog Rule

These package names and profiles are reference offerings, not guaranteed final system recommendations. The AI should use customer information and the knowledge base before suggesting a package.

9. Reference Package Guidance

Customer Signal	Potential Reference Offering	Do Not Assume
Moderate residential requirement	HomeStart 3	Do not assume 3 kW is suitable without consumption/site information.
Higher residential requirement	HomePro 5	Do not assume 5 kW is sufficient for every household.
Large residential requirement	HomeMax 10	Do not assume rooftop suitability.
Small commercial requirement	Business 15	Do not assume commercial load or rooftop suitability.
Larger small-business requirement	Business 25	Do not assume final system size.

10. Reference Service Locations

For this assessment, the following locations are within Vantis's reference service footprint. The list is intentionally limited; locations not listed should be treated as unverified.

Region	Reference Locations	Status
Uttar Pradesh	Lucknow, Prayagraj, Kanpur, Varanasi, Noida, Ghaziabad	Reference service area
Delhi NCR	Delhi, Gurugram, Noida, Ghaziabad, Faridabad	Reference service area
Punjab	Ludhiana, Amritsar, Jalandhar	Reference service area
Haryana	Panipat, Karnal, Sonipat	Reference service area
Rajasthan	Jaipur, Kota	Reference service area

If a customer mentions a city not appearing above, the AI should say that service availability is not confirmed in its current information and offer to collect details for verification.

11. Reference Warranty & Support Policy

Component / Area	Reference Policy
Solar Panels	Warranty is governed by the applicable manufacturer and project documentation. No universal duration should be assumed.
Inverter	Warranty depends on the selected equipment and manufacturer terms.
Installation Work	Applicable coverage depends on the project agreement and installation scope.
Post-Installation Support	Vantis can assist with maintenance and support requests; exact scope depends on project/equipment terms.
Existing Customer Issue	Collect customer name, contact information, project details and issue description before escalating.

12. Reference Support Scenarios

Scenario	AI Response Direction
Customer reports inverter issue	Collect identity/contact + project details + issue description; create/update a support request if the system supports it.
Customer asks about warranty duration	Explain that duration depends on the specific component/manufacturer terms.
Customer asks for maintenance	Explain that Vantis provides post-installation support and collect project details.
Customer asks for installation date	Do not guarantee; explain that scheduling depends on project/site/equipment factors.

13. Financing & Government Subsidies

Financing availability may depend on the project, financial institution, eligibility, and applicable terms. The AI must not promise financing approval or a specific EMI.

Government subsidy programs, eligibility requirements, and amounts may change over time. The AI must not state a specific subsidy amount unless explicitly present in this knowledge base. Customers should be told that current eligibility and benefits require verification.

14. Customer Information for Consultation

Basic Information

- Full name
- Phone number
- Email address

Property Information

- City
- State
- Property type
- Ownership status, if relevant
- Rooftop availability

Energy Requirement

- Approximate monthly electricity bill
- Approximate electricity consumption

- Existing solar system, if any
- Desired solar system, if already known

Purchase Intent

- Reason for considering solar
- Expected installation timeline
- Budget, if the customer is comfortable sharing
- Specific requirements

The AI should collect information naturally rather than asking every question at once.

15. Frequently Asked Questions

What does Vantis Energy Solutions do?

Vantis provides rooftop solar consultation, installation, and related energy solutions for residential and small commercial customers.

Do you install solar panels?

Yes. Vantis provides solar installation services for eligible residential and commercial projects.

Can you help me choose a solar system?

Yes. The team can evaluate electricity consumption, monthly bill, property type, rooftop availability, location, and requirements before recommending a suitable solution.

How much does solar installation cost?

There is no single fixed price. Cost depends on system size, equipment, property, installation requirements, location, and project complexity.

How much can I save?

Savings depend on consumption, system configuration, electricity tariffs, location, and project-specific factors. The AI should not promise a specific savings amount.

Do you provide financing?

Options may be available depending on the project and eligibility. A representative can provide more information.

Do you provide subsidy assistance?

The team can explain applicable considerations, but current eligibility and subsidy amounts require verification.

What warranty do solar panels have?

Warranty coverage depends on the specific equipment and manufacturer terms.

How long does installation take?

Timing varies with project, site conditions, equipment availability, documentation, and scheduling.

Do you provide maintenance?

Yes. Vantis provides post-installation support and maintenance-related assistance.

Can I get a quote?

Yes. Customers can provide contact and project details so the sales team can evaluate the requirement.

16. Information the AI Must Not Invent

- Exact solar system price
- Exact savings amount
- Exact subsidy amount
- Guaranteed subsidy eligibility
- Guaranteed installation date

- Exact warranty duration when equipment is unspecified
- Service availability in an unverified location
- Financing approval
- EMI amount
- Equipment specifications not present in this document
- Appointments that were not actually scheduled
- Human representative actions that did not occur

17. AI Behavior Guidelines

- Be conversational
- Be concise
- Ask relevant follow-up questions
- Use the knowledge base as the primary source of truth
- Identify buying intent
- Collect lead information naturally
- Avoid unnecessary questions
- Explain uncertainty
- Avoid hallucination
- Recommend appropriate next actions

The assistant should **not behave like an aggressive salesperson**.

18. Source-of-Truth Rule

This document is the primary source of truth for Vantis Energy Solutions for the assignment.

Intended information flow: **Knowledge Base** → **Retrieved Context** → **AI Response**.

The AI should prioritize retrieved knowledge over assumptions or general model knowledge.

If an answer cannot be supported: do not hallucinate; communicate the limitation; ask for relevant information if needed; and offer sales-team follow-up where appropriate.

19. RAG Evaluation Hints

The product, service-area, warranty, and support sections contain intentionally specific facts. A good RAG implementation should retrieve the correct supporting information instead of relying on generic solar knowledge.

Candidates are expected to make their own decisions about chunking, embeddings, metadata, retrieval strategy, reranking, context construction, and citation/source attribution. This document does not prescribe an implementation.